

Leveraging the Pre-trained Whisper Encoder for Multilingual Language Identification

1st Yunus Emre Ozkose

Sestek

Istanbul, Turkey

yunusemre.ozkose@sestek.com

2nd Ali Haznedaroglu

Sestek

Istanbul, Turkey

ali.haznedaroglu@sestek.com

Abstract—Language Identification (LID) is a critical component of modern speech processing systems. Recent advances in large-scale, pre-trained speech models like OpenAI’s Whisper offer new paradigms for downstream tasks, yet their direct applicability to LID remains underexplored. This paper evaluates the efficacy of the Whisper model for language identification, comparing its performance against a strong, specialized CNN-based baseline (EfficientAT-DyMN). Our experiments are conducted across two distinct domains: narrow-band telephony audio and wide-band Common Voice (CV) data, with an additional focus on robustness to accented speech. Our results reveal that pre-trained Whisper models, when used as simple feature extractors, perform poorly due to a fundamental task misalignment with their ASR pre-training objective. However, through end-to-end fine-tuning, the Whisper-tiny model’s performance is dramatically enhanced. It achieves a state-of-the-art accuracy of 89.70% on the challenging telephony dataset, significantly surpassing the baseline’s 75.20%. Furthermore, the fine-tuned model demonstrates exceptional robustness to accent variations, outperforming the baseline by 16-17 absolute percentage points on American-accented Spanish and French. We conclude that while large pre-trained models are not effective “out-of-the-box” for LID, the strategy of fine-tuning unlocks their potential, establishing a new and highly effective approach for creating accurate and robust LID systems capable of handling diverse, real-world acoustic conditions.

Index Terms—Spoken Language Identification (LID), Multilingual Audio Processing, Callcenter Domain Dataset

I. INTRODUCTION

In an increasingly interconnected world, the volume of digital audio content is expanding at an unprecedented rate. This surge in multilingual data from sources like social media, global broadcasting, and user-generated platforms has created a critical need for technologies that can automatically process and organize spoken language. Spoken Language Identification (LID), the task of automatically determining the language present in an audio clip, serves as a fundamental front-end technology. Its applications are vital and diverse, ranging from routing calls in international customer service centers and indexing vast multimedia archives to enabling downstream tasks like language-specific Automatic Speech Recognition (ASR) or machine translation. However, building robust LID systems is inherently challenging, as they must contend with immense variability in acoustic environments, channel distortions (e.g., telephony versus clean studio recordings), diverse speaker accents, and varying utterance lengths.

The evolution of LID has been marked by a transition from traditional statistical methods, such as Gaussian Mixture Models [1], to sophisticated deep learning architectures that can learn feature hierarchies directly from data [2]. More recently, the field has been revolutionized by large-scale,

self-supervised learning (SSL) models pre-trained on vast quantities of unlabeled audio. These models learn powerful, universal speech representations that can be effectively transferred to various downstream tasks. Among these, OpenAI’s Whisper model represents a paradigm shift, distinguished by the unprecedented scale and diversity of its training corpus—over 680,000 hours of multilingual and multitask supervised data from the web [15]. While Whisper was designed as a complete encoder-decoder system for ASR and speech translation, its encoder is implicitly trained to extract rich, discriminative features that are remarkably robust to a wide array of speakers, accents, and acoustic conditions.

This architectural design, however, presents both an opportunity and a challenge. The full Whisper model, with its generative decoder, is fundamentally over-specified for a purely discriminative task like LID, which only requires a single label output for an entire utterance. Relying on the full architecture introduces unnecessary computational overhead, increases inference latency, and adds a layer of complexity that is not aligned with the task’s objective. This leads to a critical research question: can the immense representational power of the Whisper model be distilled into a more targeted and efficient architecture for language identification, and can such a model outperform specialized, state-of-the-art systems?

In this paper, we answer this question affirmatively by proposing and evaluating a streamlined yet powerful approach for spoken language identification. Our methodology involves fine-tuning the pre-trained Whisper encoder while excising the decoder component entirely, appending a simple linear classification layer directly onto the encoder’s output. We hypothesize that this lightweight, encoder-only architecture not only retains but excels at the LID task by leveraging the robust features learned during pre-training. To validate this, we conduct a rigorous empirical evaluation against a strong, specialized baseline model across multiple demanding audio domains, including narrow-band telephony data, wide-band Common Voice data, and specific subsets of accented speech.

Our main contributions are threefold: 1) We demonstrate that fine-tuning the Whisper encoder is a highly effective strategy for language identification, establishing a new state-of-the-art performance, particularly on challenging, real-world telephony data. 2) We provide a comprehensive analysis of the model’s robustness, showing that the diverse pre-training of Whisper leads to superior generalization on accented speech compared to specialized baseline models. 3) We prove that the generative components of the Whisper

architecture are unnecessary for this discriminative task, presenting a streamlined model that achieves competitive performance while significantly reducing architectural complexity and computational requirements.

The remainder of this paper is organized as follows: Section 2 reviews related works in language identification. Section 3 details our proposed methodology. Section 4 describes our experimental setup and presents the results. Finally, Section 6 concludes the paper and suggests avenues for future work.

II. RELATED WORKS

The field of Spoken Language Identification (LID) has evolved significantly over the past decades. Early approaches were dominated by statistical methods that modeled either the acoustic or the phonotactic characteristics of a language. Acoustic models, such as Gaussian Mixture Models (GMMs), were used to map the distribution of features like Mel-Frequency Cepstral Coefficients (MFCCs). This paradigm culminated in the i-vector framework, which created a compact, fixed-dimensional representation of an audio utterance's total variability, proving highly effective when classified with SVMs or PLDA [4]. Complementing this, phonotactic approaches like Phone Recognition followed by Language Modeling (PR-LM) first converted audio into a sequence of phone-like units and then used n-gram models to score the likelihood of that sequence for each target language [19]. These traditional systems, while powerful, relied on hand-crafted features and complex, multi-stage pipelines.

The advent of deep learning initiated a shift towards data-driven, end-to-end systems. Initially, Deep Neural Networks (DNNs) were used to improve upon traditional frameworks, for example, by replacing the final classifier for i-vectors. A pivotal development was the x-vector architecture, which employed a Time Delay Neural Network (TDNN) to learn a discriminative utterance-level embedding directly from acoustic features [17]. This marked a move towards architectures that could learn hierarchical features automatically. Fully end-to-end models soon followed, using Convolutional Neural Networks (CNNs) to capture local spectro-temporal patterns from spectrograms and Recurrent Neural Networks (RNNs) to model long-range temporal dependencies. Hybrid architectures combining CNNs and RNNs became particularly popular, demonstrating strong performance by leveraging the complementary strengths of both [7], [9].

Current state-of-the-art LID is largely defined by two innovations: attention mechanisms and self-supervised learning. The attention mechanism, popularized by the Transformer architecture, allows a model to dynamically weigh the importance of different audio frames, focusing on the most linguistically informative segments and improving robustness to noise and silence [10]. The most significant breakthrough, however, has been the adoption of large-scale, self-supervised learning (SSL) models like Wav2Vec 2.0 [2], HuBERT [8], and WavLM [3]. These models are pre-trained on thousands of hours of unlabeled audio to learn powerful and universal speech representations. By simply fine-tuning these pre-trained models on a small amount of labeled data, researchers have achieved unprecedented accuracy in LID, drastically outperforming systems trained from scratch and setting a new standard for performance, particularly in low-resource scenarios.

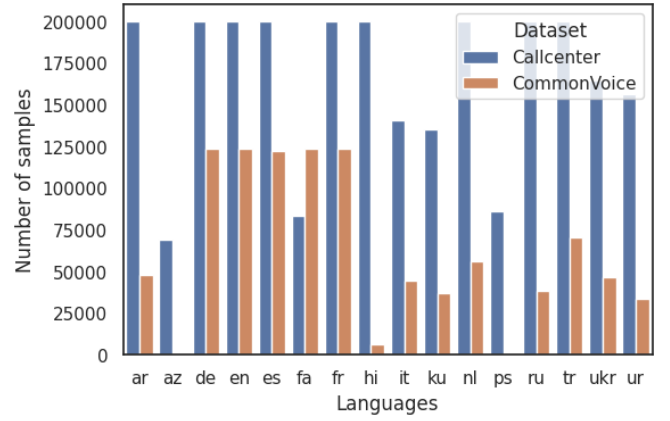


Fig. 1. Per-language sample distribution for the training data. This chart compares the number of samples sourced from our internal Callcenter dataset (blue) and the public Common Voice dataset (orange). The Callcenter set provides a substantial volume of data for most languages, while the Common Voice set exhibits greater variability in sample counts across the 16 languages.

Several studies have addressed audio classification tasks in call center environments, focusing on segmenting and categorizing call recordings to enhance customer relationship management and operational efficiency. Ozan [12] proposed a convolutional recurrent neural network (CRNN) framework to accurately classify audio segments as either customer or agent speech, demonstrating strong generalization across languages despite training on Turkish data. Park et al. [13] explored call-type classification using partial transcripts, revealing that analyzing only the first 40 speaker turns or solely the agent's utterances can achieve classification accuracy comparable to full-call transcripts, significantly reducing transcription costs. Robust automatic speech recognition (ASR) systems tailored for noisy call center conditions have also been developed, with LSTM-based architectures showing superior performance in handling varying noise levels, thereby improving transcription quality essential for downstream analytics by [14]. Additionally, emotion recognition in call center voice data has gained attention, with recent approaches leveraging self-supervised learning models to detect emotional states and improve service protocols in industrial applications [11]. The study conducted by [6] employs ECAPA-TDNN [5] and X-Vector [18] models to perform language identification across six languages within the call center domain. Collectively, these works highlight the integration of neural architectures, partial transcript utilization, and advanced speech and emotion recognition techniques as key components in advancing audio language classification and analytics in call center scenarios.

III. APPROACH

A. Data

To train and evaluate our language identification model, we constructed a comprehensive dataset covering 16 distinct languages. This dataset is a composite of two primary sources: a large-scale internal corpus, which we refer to as the Callcenter dataset, and the publicly available Mozilla Common Voice (CV) dataset [1]. This approach enhances the model's robustness for processing speech captured via both telephony and microphone channels. The datasets comprise the following languages: Arabic (ar), Azerbaijani (az),

German (de), English (en), Spanish (es), Persian (Farsi) (fa), French (fr), Hindi (hi), Italian (it), Kurdish (ku), Dutch (nl), Pashto (ps), Russian (ru), Turkish (tr), Ukrainian (ukr), and Urdu (ur).

As illustrated in Fig. 2, the contribution of each source varies significantly across languages. The Callcenter dataset provides a substantial and relatively uniform volume of data for most languages, forming the backbone of our training corpus. The Common Voice dataset supplements this, though its sample counts are more varied, leading to a notable class imbalance in the final aggregated dataset. Before use, all audio files were pre-processed and standardized to a 16 kHz sampling rate and a single-channel (mono) format. The audio segment duration was constrained to 5 seconds through the application of truncation and repetition padding techniques.

B. Model

We implemented and compared two distinct model architectures for the LID task, each leveraging a powerful pre-trained audio foundation. Our first model is based on the encoder of the OpenAI Whisper architecture (whisper-tiny). Whisper is a Transformer-based model pre-trained on 680,000 hours of multilingual and multitask supervised speech data. Its encoder is specifically designed to learn rich, contextualized representations of speech. Our second model leverages Dymn-10 [16], a Dynamic Convolutional Neural Network pre-trained on the large-scale AudioSet dataset. Unlike Whisper, AudioSet contains a diverse range of general audio events (e.g., music, animal sounds, environmental noises) in addition to human speech. This allows us to test the hypothesis that a general audio representation can be as effective, or more effective, than a purely speech-focused representation for LID.

The Whisper model typically consists of multiple encoder-decoder blocks designed for tasks like speech recognition and translation. For the purpose of effective language identification, we introduced the following modifications: 1) removal of decoder blocks and 2) addition of a fully connected layer. Since language identification is essentially a classification task rather than a generative task, we opted to remove the decoder blocks from the Whisper architecture. This modification helps in reducing computational complexity and focuses the model on feature extraction relevant to language classification. Number of parameter is decreased by 78.23% in that way. To adapt the model for classifying languages, a fully connected (FC) layer was introduced at the end of the encoder. This layer consists of 16 output neurons, each corresponding to one of the languages in our dataset. The FC layer is vital for mapping the encoder’s output to probabilistic predictions for each language. Fig. 2 shows model architecture. Similar to the Whisper-based model, we appended a classification head composed of a fully connected (FC) layer with 16 output neurons and a subsequent Softmax activation function.

IV. EXPERIMENTS

A. Experimental Setup

The model was trained using supervised learning with a cross-entropy loss function, which is well-suited for multi-class classification tasks like language identification. We employed the Adam optimizer due to its efficiency and adaptive learning rate capabilities. The model’s performance

was evaluated using standard metrics for classification tasks, including accuracy, precision, recall, and F1-score. Each metric was computed per language to ensure an in-depth analysis of the model’s strengths and weaknesses across different languages. Additionally, confusion matrices were constructed to visualize the model’s performance in distinguishing closely-related languages.

The dymn10 model establishes a formidable baseline, achieving 75.20% accuracy on the telephony dataset and an impressive 90.80% on the Common Voice (CV) dataset. As a specialized audio CNN, its high performance on the CV dataset is expected, effectively capturing the rich spectro-temporal features from the clean, wide-band audio. The performance drop on the telephony dataset highlights the inherent difficulty of this domain, which is characterized by narrow-band (8kHz) audio, channel noise, and conversational speech patterns. Even a specialized architecture like DyMN10 finds the degraded signal challenging.

A striking observation is the poor performance of the pre-trained Whisper models when used as simple feature extractors. Both Whisper-tiny-pretrained (43.10% Telephony, 70.00% CV) and Whisper-base-pretrained (46.60% Telephony, 77.20% CV) perform significantly worse than the dymn10 baseline. This result underscores a crucial task misalignment. Whisper’s pre-training on automatic speech recognition (ASR) and translation optimizes its internal representations for generating textual output, not for the high-level discriminative task of identifying the language itself. Without adaptation, these powerful features are not directly suitable for language classification. Between the two pre-trained Whisper variants, the larger Whisper-base model shows a modest improvement over Whisper-tiny on both datasets. This suggests that larger model capacity provides more generalizable features, though still insufficient for high performance without task-specific tuning.

The most significant finding of our study is the dramatic performance boost achieved through fine-tuning. The Whisper-tiny-finetuned model achieves an accuracy of 89.70% on the telephony dataset and 80.20% on the CV dataset. This result demonstrates the critical importance of adapting the pre-trained model’s weights to the target task and domain. On the challenging telephony dataset, the fine-tuned model not only overcomes its poor initial performance but also surpasses the strong dymn10 baseline by over 14 absolute percentage points. This is a remarkable achievement, indicating that the transformer-based architecture, once adapted, is exceptionally effective at learning robust representations from noisy, narrow-band audio. It likely captures long-range acoustic dependencies and contextual cues that even a specialized CNN struggles with.

Interestingly, the Whisper-tiny-finetuned model exhibits a performance pattern that is the reverse of dymn10: it performs better on the telephony data (89.70%) than on the CV data (80.20%). While its performance on CV is strong, it does not reach the level of the specialized DyMN10 model. This reversal suggests that the fine-tuning process allowed the Whisper model to specialize profoundly for the most challenging characteristics of the telephony domain. The model learned to be robust to noise, channel effects, and limited bandwidth, making it the state-of-the-art performer for that specific condition in our experiments. In contrast, the DyMN10 architecture maintains its edge on cleaner, more

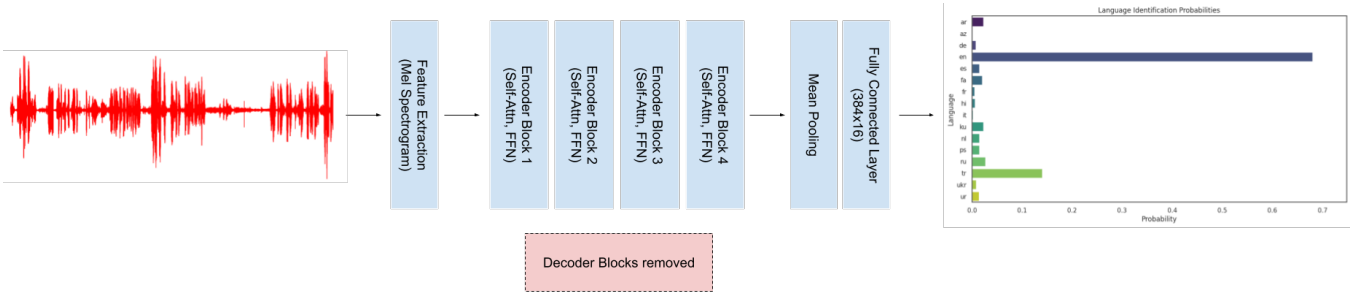


Fig. 2. Language identification pipeline based on the Whisper encoder. The input waveform is first transformed into a Mel spectrogram and passed through the Whisper encoder blocks. Decoder blocks are removed, and the encoder output is mean-pooled and fed into a fully connected layer for classification. The resulting language probabilities are shown on the right, with the highest prediction corresponding to the correct language.

TABLE I
COMPARATIVE PERFORMANCE OF VARIOUS MODELS ON LANGUAGE IDENTIFICATION TASKS, EVALUATED ACROSS TWO DATASETS: TELEPHONY AND COMMON VOICE (CV). THE TABLE PRESENTS THE ACCURACY RATES ACHIEVED BY EACH MODEL VARIANT, HIGHLIGHTING THE IMPACT OF FINE-TUNING ON THE WHISPER ARCHITECTURE. THE WHISPER-TINY-FINETUNED MODEL SHOWS SIGNIFICANT IMPROVEMENTS IN ACCURACY FOR BOTH DATASETS, DEMONSTRATING THE BENEFITS OF DOMAIN-SPECIFIC TRAINING.

Model	Telephony Accuracy	CV Accuracy
EfficientAT-DyMN	75.20	90.80
Whisper-tiny-pretrained	43.10	70.00
Whisper-tiny-finetuned	89.70	80.20
Whisper-base-pretrained	46.60	77.20

conventional audio data.

To further probe the models’ generalization capabilities and assess their performance in more challenging, real-world scenarios, we conducted a targeted evaluation on accented speech. For this purpose, we curated a specific test subset containing audio clips of native American English speakers speaking Spanish and French. The primary objective of this experiment was to measure the models’ ability to correctly identify the language being spoken (the target L2: Spanish or French) without being confounded by the speaker’s L1 accent (American English). This stress test is designed to reveal whether the models have learned to identify fundamental linguistic features of a language (e.g., phonology, cadence, vocabulary) or if they are overly reliant on the prototypical accent profiles learned from the training data. The analysis focuses on quantifying misclassifications where the predicted language is the speaker’s native English, rather than the language being spoken. The test dataset comprises 100 samples for each language from Common Voice. Experiments were conducted specifically on American-accented Spanish and American-accented French. The results of these experiments are presented in Table II.

The Whisper-tiny-finetuned model demonstrates a commanding advantage on both accented language datasets. It achieves 96.00% accuracy on American Spanish and 82.00% on American French. In contrast, the EfficientAT-DyMN model, while a strong baseline, achieves only 79.00% and 66.00%, respectively. This represents a substantial performance margin of 17 percentage points for Spanish and 16 percentage points for French. We attribute this superior robustness primarily to the nature of Whisper’s pre-training. The model was trained on 680,000 hours of audio from the web, a dataset of immense diversity encompassing a

vast array of speakers, accents, acoustic environments, and speaking styles. This exposure has endowed the model with highly generalized representations of speech that are less sensitive to phonetic deviations from a “standard” accent. Consequently, when fine-tuned, the model can more easily associate accented phonology with the correct base language. The EfficientAT-DyMN model, like many CNN-based architectures, excels at learning discriminative spectro-temporal patterns. However, its performance on accented speech suggests that it may have learned patterns more specific to the “standard” accents present in its training data. When faced with the phonetic variations of American-accented Spanish and French, the model’s learned filters may not generalize as effectively, leading to a higher error rate. The performance drop, especially on French (66.00%), indicates a significant challenge in distinguishing the accented language from others. Beyond pre-training, the transformer architecture of Whisper may also contribute to its robustness. The self-attention mechanism allows the model to weigh the importance of different acoustic features across an entire utterance. This global context may help the model look past localized phonetic variations (e.g., a single vowel sound) and focus on broader, more stable linguistic cues like rhythm, intonation, and syntax encoded in the audio, which remain more consistent across accents.

The confusion matrix in Fig. 3 demonstrates the performance of a Whisper model fine-tuned on 16 languages using Common Voice and call center data, evaluated exclusively on call center speech. The model exhibits strong diagonal dominance, indicating high classification accuracy across most languages, with particularly strong results for Turkish (70/72), German (64/64), and Persian (63/64). However, certain language pairs show notable confusion, especially Urdu, which is often misclassified as Arabic, Hindi, or Turkish due to phonetic and lexical overlap; and Ukrainian-Russian, which exhibit mutual confusion (20 and 12 samples, respectively), likely due to their linguistic proximity. Spanish shows minor confusion with English and Portuguese, reflecting intonational similarities and possible code-switching. Russian and Urdu display the lowest accuracy, with significant cross-class mispredictions, suggesting either insufficient or acoustically ambiguous data. Overall, errors predominantly occur between linguistically or geographically related languages, indicating the influence of shared phonological traits and domain-specific variability in call center speech. These findings suggest potential gains from contrastive learning, phoneme-level modeling, or increased domain adaptation for

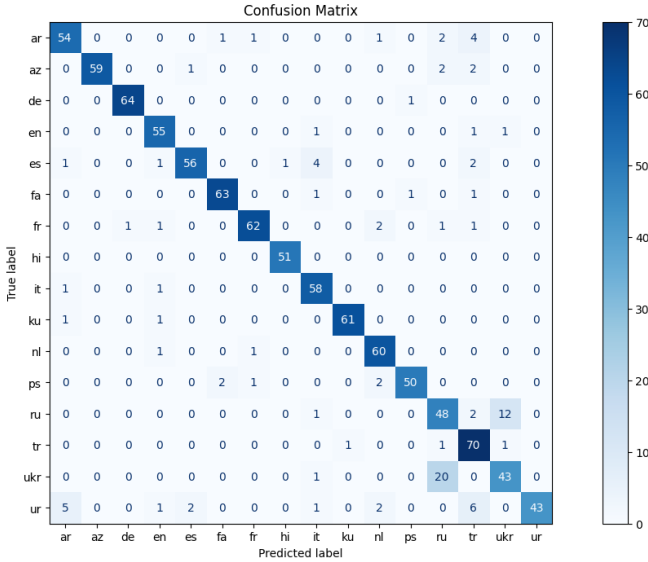


Fig. 3. Confusion matrix of the Whisper model fine-tuned on 16 languages (Common Voice + call center) and evaluated on call center test data. The matrix shows strong classification accuracy for most languages, with notable confusions between linguistically similar pairs such as Ukrainian–Russian and Urdu–Arabic/Hindi. These errors highlight challenges in disambiguating closely related languages in real-world call center speech.

TABLE II
COMPARATIVE PERFORMANCE ON ACCENTED LANGUAGE IDENTIFICATION. THE RESULTS DEMONSTRATE THE SUPERIOR ROBUSTNESS OF THE WHISPER-TINY-FINETUNED MODEL OVER THE EFFICIENTAT-DYMN BASELINE WHEN CLASSIFYING AMERICAN-ACCENTED SPANISH AND FRENCH.

Model	Spanish	French
EfficientAT-DyMN	79.00	66.00
Whisper-tiny-finetuned	96.00	82.00

closely related language pairs.

The confusion matrix in Fig. 3 presents the performance of the Whisper model fine-tuned on 16 languages using combined Common Voice and call center data, evaluated on the Common Voice test set. The model achieves high classification accuracy across most languages, with Spanish (86), Dutch (82), and Italian (76) showing the strongest results. Compared to call center evaluations, confusion is generally lower, likely due to the more controlled and cleaner recording conditions in Common Voice. However, some misclassifications persist, particularly between phonetically or typologically similar languages—such as English with Spanish and Italian, Urdu with Pashto and Hindi, and Ukrainian with Russian—indicating challenges in disambiguating related languages even in read speech. The higher performance overall suggests that the model effectively learns robust language representations but may still benefit from additional disambiguation strategies, particularly for low-resource or closely related language pairs.

V. CONCLUSION

This work investigated the efficacy of applying large-scale, pre-trained speech models, specifically OpenAI’s Whisper, to the task of language identification across diverse and challenging audio domains. By comparing its performance against a strong, specialized baseline (EfficientAT-DyMN),

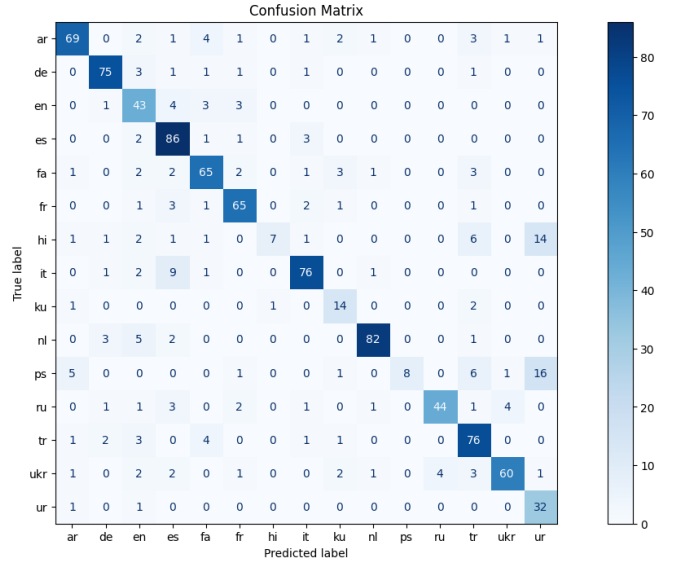


Fig. 4. Confusion matrix of the Whisper model fine-tuned on 16 languages evaluated on Common Voice test data. The model demonstrates strong performance on most languages, with particularly high accuracy for Spanish (86), Dutch (82), and Italian (76). Notable confusions occur between English and Germanic/Romance languages (e.g., English–Spanish/Italian), as well as between Urdu and phonetically similar South Asian languages such as Hindi and Pashto. Ukrainian and Russian again show moderate confusion, highlighting persistent challenges in separating closely related languages. These results confirm the model’s generalization ability on read speech while reflecting known cross-linguistic phonetic overlaps.

we have drawn several key conclusions that carry significant implications for the field.

Our primary finding is that the utility of a foundation model like Whisper is entirely dependent on adaptation. When used as an “off-the-shelf” feature extractor, the model’s performance was strikingly poor, lagging significantly behind the specialized CNN baseline. This highlights a crucial task misalignment: the internal representations learned for Automatic Speech Recognition (ASR) are not inherently optimized for the discriminative classification task of language identification.

However, the process of end-to-end fine-tuning proved to be transformative. The Whisper-tiny-finetuned model not only reversed its initial deficit but established a new state-of-the-art performance on the most challenging dataset—the narrow-band telephony audio—surpassing the EfficientAT-DyMN baseline by over 14 percentage points. This demonstrates that the rich, general-purpose features learned during pre-training can be effectively molded to excel at a specific downstream task, unlocking the model’s true potential.

Furthermore, our experiments on accented speech underscored the profound robustness of the fine-tuned Whisper model. It outperformed the baseline by a remarkable margin (16-17 percentage points) when classifying American-accented Spanish and French. We attribute this superior generalization to the immense diversity of its pre-training data, which has endowed the model with an intrinsic ability to handle phonetic and speaker variations that often challenge more specialized architectures.

In summary, this study demonstrates that the path to building highly robust and accurate language identification systems is shifting. While specialized architectures remain effective, particularly on clean data, the strategy of adapt-

ing large, pre-trained foundation models through fine-tuning offers a more powerful and generalizable solution. These models are not only capable of achieving state-of-the-art accuracy but also exhibit superior robustness to real-world complexities like channel degradation and accent variation.

Future work should extend this analysis by fine-tuning larger variants of the Whisper family (e.g., base, small) to assess the impact of model scale on final performance. Additionally, exploring the data efficiency of the fine-tuning process and expanding the evaluation to a wider array of languages and accents would provide a more complete picture of the capabilities of this promising approach.

REFERENCES

- [1] Ardila, R., Branson, M., Davis, K., Henretty, M., Kohler, M., Meyer, J., Morais, R., Saunders, L., Tyers, F.M., Weber, G.: Common voice: A massively-multilingual speech corpus (2020), <https://arxiv.org/abs/1912.06670>
- [2] Baevski, A., Zhou, H., Mohamed, A.R., Auli, M.: wav2vec 2.0: A framework for self-supervised learning of speech representations. In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. pp. 12449–12460 (2020)
- [3] Chen, S., Chen, C., Wu, T., Wang, Y., Chen, S.Y., Kwok, D., Wang, S., Zhang, Q., Gan, Z., Li, C., Li, X., Zeng, X., Liu, S., Zhang, J., Zeng, M., Wang, W., Zhou, Y., Wei, F.: Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **30**, 2371–2384 (2022). <https://doi.org/10.1109/TASLP.2022.3188521>
- [4] Dehak, N., Kenny, P.J., Dehak, R., Dumouchel, P., Ouellet, P.: Front-end factor analysis for speaker verification. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **19**(4), 788–798 (2011). <https://doi.org/10.1109/TASL.2010.2064307>
- [5] Desplanques, B., Thienpondt, J., Demuynck, K.: Ecapa-tdnn: Emphasized channel attention, propagation and aggregation in tdnn based speaker verification. *arXiv preprint arXiv:2005.07143* (2020)
- [6] Furkan Çiğdem, E., Haznedaroğlu, A., Arslan, L.M.: Spoken language identification using call center data. In: *2021 Innovations in Intelligent Systems and Applications Conference (ASYU)*. pp. 1–4 (2021). <https://doi.org/10.1109/ASYU52992.2021.9599040>
- [7] Gonzalez-Dominguez, J., Lopez-Moreno, I., Sak, H., Gonzalez-Rodriguez, J., Moreno, P.J.: Automatic language identification using long short-term memory recurrent neural networks. In: *Interspeech*. pp. 2155–2159 (2014)
- [8] Hsu, W.N., Bolte, B., Tsai, Y.H.H., Lakhotia, K., Salakhutdinov, R., Mohamed, A.: Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM transactions on audio, speech, and language processing* **29**, 3451–3460 (2021)
- [9] Lopez-Moreno, I., Gonzalez-Dominguez, J., Plchot, O., Martinez, D., Gonzalez-Rodriguez, J., Moreno, P.: Automatic language identification using deep neural networks. In: *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. pp. 5337–5341 (2014). <https://doi.org/10.1109/ICASSP.2014.6854622>
- [10] Lozano-Diez, A., Zazo, R., Gonzalez-Rodriguez, J., Toledano, D.T.: On the use of self-attention for spoken language identification. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **30**, 1969–1981 (2022). <https://doi.org/10.1109/TASLP.2022.3180299>
- [11] Martín-Doñas, J.M., Zorrilla, A.L., deVelasco, M., Vázquez-Correa, J.C., Álvarez, A., Torres, M.I., Delgado, P., Lazpiur, A., Romero, B., Alkorta, I.: Speech emotion recognition for call centers using self-supervised models: A complete pipeline for industrial applications. In: *Proceedings of the 7th International Conference on Natural Language and Speech Processing (ICNLSP 2024)*. pp. 119–128 (2024)
- [12] Şükrü Ozan: Classification of audio segments in call center recordings using convolutional recurrent neural networks (2021), <https://arxiv.org/abs/2106.02422>
- [13] Park, Y., Teiken, W., Gates, S.C.: Low-cost call type classification for contact center calls using partial transcripts. In: *INTERSPEECH*. pp. 2739–2742 (2009)
- [14] Parra-Gallego, L.F., Arias-Vergara, T., Orozco-Arroyave, J.R.: Robust automatic speech recognition for call center applications. In: *WEA*. pp. 72–83 (2021), https://doi.org/10.1007/978-3-030-86702-7_7
- [15] Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: *International conference on machine learning*. pp. 28492–28518. PMLR (2023)
- [16] Schmid, F., Koutini, K., Widmer, G.: Dynamic convolutional neural networks as efficient pre-trained audio models. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2024)
- [17] Snyder, D., Garcia-Romero, D., Sell, G., Povey, D., Khudanpur, S.: X-vectors: Robust dnn embeddings for speaker recognition. In: *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. pp. 5329–5333 (2018). <https://doi.org/10.1109/ICASSP.2018.8461375>
- [18] Snyder, D., Garcia-Romero, D., Sell, G., Povey, D., Khudanpur, S.: X-vectors: Robust dnn embeddings for speaker recognition. In: *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. pp. 5329–5333. IEEE (2018)
- [19] Zissman, M.A.: Comparison of four approaches to automatic language identification of telephone speech. *IEEE Communications Magazine* **34**(10), 84–91 (1996). <https://doi.org/10.1109/35.544331>